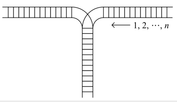
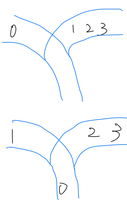
100

1. (10%) Consider the railroad switching network. Railroad cars numbered 0, 1, 2, …, n-1 are at the right. Each car is brought into the stack and removed at any time. For instance, if $n = 3$, we could move in 0, move in 1, move in 2, and then take the cars out, producing the new order 2, 1, 0. Find all possible permutations for $n = 4$.



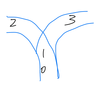


0 1 2 3
0 1 3 2
0 2 1 3
0 2 3 1
0 3 2 1

1 0 2 3
1 0 3 2
1 2 0 3
1 2 3 0
1 3 2 0

2 1 0 3
2 1 3 0
2 3 1 0

3 2 1 0



2. (20%) Given a **stack** of size 4, according to Program 3.1 to 3.3 in the textbook, what is the stack in each step when we do the following commands:
push J1, push J2, push J3, pop, push J4, push J5, push J6

top	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1					stack is empty
0	J_1				J_1 is pushed
1	J_1	J_2			J_2 is pushed
2	J_1	J_2	J_3		J_3 is pushed
1	J_1	J_2			J_3 is popped
2	J_1	J_2	J_4		J_4 is pushed
3	J_1	J_2	J_4	J_5	J_5 is pushed
3	J_1	J_2	J_4	J_3	stack is full

3. (20%) Given a **queue** of size 4, according to Program 3.5 to 3.6 in the textbook, what is the stack in each step when we do the following commands:
add J1, add J2, add J3, delete, add J4, add J5, add J6

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1	-1					queue is empty
-1	0	J_1				J_1 is added
-1	1	J_1	J_2			J_2 is added
-1	2	J_1	J_2	J_3		J_3 is added
0	2		J_2	J_3		J_1 is deleted
0	3		J_2	J_3	J_4	J_4 is added
0	3		J_2	J_3	J_4	queue is full
0	3		J_2	J_3	J_4	queue is full

4. (20%) Given a **circular queue** of size 4, according to Program 3.7 to 3.8 in the textbook, what is the stack in each step when we do the following commands:
add J1, add J2, add J3, delete, add J4, add J5, delete, delete

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
0	0					queue is empty
0	1		J_1			J_1 is added
0	2		J_1	J_2		J_2 is added
0	3		J_1	J_2	J_3	J_3 is added
1	3			J_2	J_3	J_1 is deleted
1	0	J_4		J_2	J_3	J_4 is added
1	0	J_4		J_2	J_3	queue is full
2	0	J_4			J_3	J_2 is deleted

3 0 J_4 J_3 is deleted

5. (20%) Write the postfix form of the following infix expressions:
(a) (A + B) * C + D / (E + F * G) - H
(b) (A && B) || C || ! (E > F)

(a)
(((A + B) * C) + (D / (E + F * G))) - H
⇒ (((AB + C * (D (EFG * + / + H -
⇒ AB + C * DEFG * + / + H - ※

(b)
(((A && B) || C) || (! (E > F)))
⇒ (((AB && C || ((EF > ! ||
⇒ AB && C || EF > ! || ※

6. (10%) Write the infix form of the following postfix expression:
A B C < C D > || ! && ! C E < ||

A
A
A
A
A
A
A
A

B
B
B<C
B<C
B<C
B<C
(B<C)||(C>D)
!((B<C)||(C>D))

C
C
C
C>D
D

A
A && !((B<C)||(C>D))
!(A && !((B<C)||(C>D)))
!(A && !((B<C)||(C>D)))
!(A && !((B<C)||(C>D)))
!(A && !((B<C)||(C>D)))

C
C
C<E
E

⇒ !(A && !((B<C)||(C>D))) || (C<E) ※

((!(A && (!((B<C) || (C>D))))) || (C<E))

A B C < C D > || ! && ! C E < ||